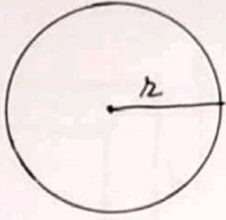
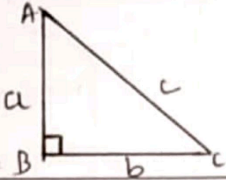
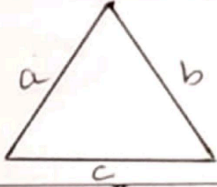
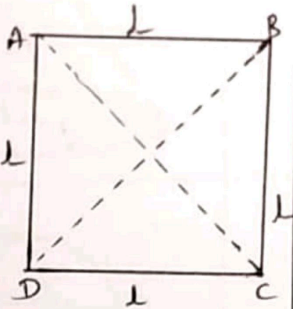
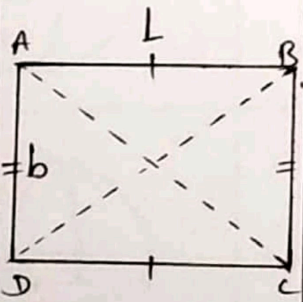
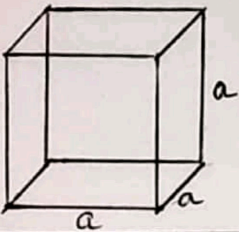
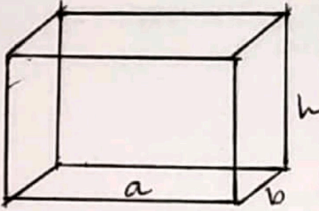
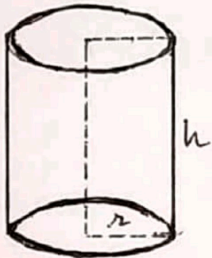
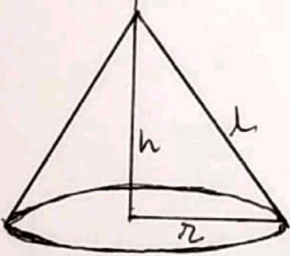
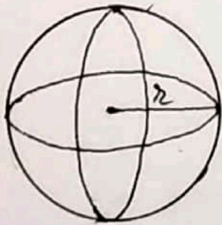
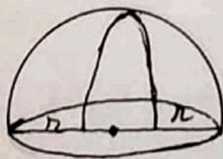


UNIT-V

MENSURATION (10 Marks)

SR. No	OBJECT	FIGURE	PERIMETER	AREA.
1	CIRCLE		$\rightarrow d = \frac{2r}{1}$ $r = \frac{d}{2}$ $2\pi r$ $\underline{= \pi d.}$	πr^2
2	TRIANGLE		$p = a + b + c$	$A = \frac{1}{2}bh$
			$p = a + b + c$	$A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$
3	SQUARE		$P = 4L$	$A = L^2$
4	RECTANGLE		$P = 2(L + b)$	$A = L \times b$

SR. NO	OBJECT	FIGURE	CURVED SURFACE AREA	TOTAL SURFACE AREA	VOLUME
1	CUBE		—	$6a^2$	a^3
2	CUBOID		—	$2(ab+bh+ah)$	abh
3	CYLINDER		$2\pi rh$	$2\pi rh + 2\pi r^2$ $= 2\pi r(h+r)$	$\pi r^2 h$
4	CONE		πrl $l = \text{slant height}$	$\pi rl + \pi r^2$ $= \pi r(l+r)$	$\frac{1}{3} \pi r^2 h$
5	SPHERE		$4\pi r^2$	$4\pi r^2$	$\frac{4}{3} \pi r^3$
6	HEMISPHERE		$2\pi r^2$	$2\pi r^2 + \pi r^2$ $= 3\pi r^2$	$\frac{2}{3} \pi r^3$

* Question for 1 Marks

1. If the Diameter of the circle is 28 cm then find its circumference.

→ $d = 28 \text{ cm}$

$$r = \frac{d}{2} = \frac{28}{2} = 14 \text{ cm}$$

$$\begin{aligned} \text{Circumference of circle} &= 2\pi r \\ &= 2 \times \frac{22}{7} \times 14^2 \\ &= 88 \text{ cm.} \end{aligned}$$

2. Find the area of square made from 120 m long wire.

→ perimeter of square is 120 m.

$$\text{perimeter of square} = 4l$$

$$\therefore 120 = 4l$$

$$\therefore l = \frac{120}{4}$$

$$\therefore \boxed{l = 30 \text{ m}}$$

$$\begin{aligned} \text{Area of square } A &= l^2 \\ &= (30)^2 \\ &= 900 \text{ m}^2. \end{aligned}$$

3. If Triangle ABC is an equilateral triangle and side $BC = 6 \text{ cm}$ then find the area of triangle ABC.

- For equilateral triangle.

$$AB = BC = AC = 6 \text{ cm.}$$

$$= a$$

$$\begin{aligned}
 \rightarrow \text{Area of equilateral triangle} &= \frac{\sqrt{3}}{4} a^2 \\
 &= \frac{\sqrt{3}}{4} (6)^2 \\
 &= \frac{\sqrt{3}}{4} \overset{9}{36}
 \end{aligned}$$

$$\therefore \boxed{A = 9\sqrt{3} \text{ cm}^2}$$

4. If r is the radius of the circle then given the formula for its circumference and area.

$$\begin{aligned}
 \rightarrow C &= 2\pi r \\
 A &= \pi r^2
 \end{aligned}$$

5. Find the area of circle made from 132 m long wire.

$\rightarrow$ circumference of circle is 132 m.

$$\text{Circumference of circle} = 2\pi r$$

$$\therefore 132 = 2\pi r.$$

$$\therefore \frac{132 \times 7}{2 \times 22} = r$$

$$\therefore 21 = r$$

$$\therefore \boxed{r = 21 \text{ m.}}$$

$$\begin{aligned}
 \text{Area of circle} &= \pi r^2 \\
 &= \frac{22}{7} \times 21^2 \times 21 \\
 &= 1386 \text{ sq. cm.}
 \end{aligned}$$

6. In a Trapezium two parallel sides are 8 cm and 12 cm. and perpendicular distance between two parallel sides is 6 cm. Find the area of Trapezium.

$$\rightarrow a = 8 \text{ cm}$$

$$b = 12 \text{ cm.}$$

$$\text{And } d = 6 \text{ cm.}$$

$$\text{Area of Trapezium} = \frac{1}{2} (a+b) \times d.$$

$$= \frac{1}{2} (8+12) \times 6$$

$$= \frac{1}{2} 20^{\cancel{10}} \times 6$$

$$\boxed{A = 60 \text{ sq. cm.}}$$

7. If the length of diagonal of rhombus is 8 cm and 15 cm. Find the area of rhombus.

$$\rightarrow \text{Here } d_1 = 8 \text{ cm, } d_2 = 15 \text{ cm.}$$

$$\text{Area of Rhombus} = \frac{1}{2} d_1 d_2$$

$$= \frac{1}{2} \times \cancel{8}^4 \times 15$$

$$= 60 \text{ sq. cm.}$$

8. If the ratios of the radius of two circles are 2:3 then find the ratio of their Areas.

$$\rightarrow \text{Let } r_1 = 2 \text{ \& } r_2 = 3.$$

$$\begin{aligned} \text{Area of circle} &= \pi r_1^2, & \text{Area of circle} &= \pi r_2^2 \\ &= \pi \times 4 & &= \pi \cdot 9 \end{aligned}$$

$$\begin{aligned} \text{Ratio of Area of circle} &= \frac{\pi r_1^2}{\pi r_2^2} \\ &= \frac{4}{9} \end{aligned}$$

Ratio of Area of circle is 4:9

9. Find the volume of a hemisphere having radius 2 meter.

→ $r = 2$ meter.

$$\begin{aligned}\text{Volume of hemisphere} &= \frac{2}{3} \pi r^3 \\ &= \frac{2}{3} \times \pi \times 8 \\ &= \frac{2}{3} \times \frac{22}{7} \times 8 \\ &= \frac{352}{21} \\ &= 16.8 \text{ m}^3.\end{aligned}$$

10. The radius of the sphere is 10 cm and its height is 14 cm then find its area of its curved surface.

→ Here $r = 10$ cm
 $h = 14$ cm.

$$\begin{aligned}\text{Curved surface area of sphere} &= 4\pi r^2 \\ &= 4 \times \frac{22}{7}\end{aligned}$$

Note:- Question is wrong. So example canele.

11. The radius of the base of cylinder is 10 cm and their heights are 14 cm. then find area of its curved surface.

→ Here $r = 10$ cm, $h = 14$ cm.

curved surface area of cylinder

$$= 2\pi r(h+r)$$

$$= 2 \times \frac{22}{7} \times 10 (14+10)$$

$$= 2 \times \frac{22}{7} \times 10 (24)$$

$$= \frac{2 \times 22 \times 240}{7}$$

$$= \frac{44 \times 240}{7}$$

$$= 1508.57 \text{ cm}^2$$

12. Give formula of volume of hemisphere.

$$V = \frac{2}{3} \pi r^3$$

13. Give formula of surface area of cube

$$S = 6l^2$$

* Question for 3 Marks

1. If measurement of sides of triangle are 7, 8 & 9 cm then find its area.

→ Let $a = 7$, $b = 8$, $c = 9$.

Area of Triangle

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\text{But } s = \frac{a+b+c}{2}$$

$$= \frac{7+8+9}{2} = \frac{24}{2} = 12 \text{ cm}$$

$$\therefore \boxed{s = 12 \text{ cm}}$$

$$\text{Now, } A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{12(12-7)(12-8)(12-9)}$$

$$= \sqrt{12 \times 5 \times 4 \times 3}$$

$$= \sqrt{12 \times 12 \times 5}$$

$$\boxed{A = 12\sqrt{5} \text{ cm}^2}$$

2. Find area of isosceles triangle whose same sides are of length 13 cm and base is of length of 24 cm.

→ Here $AB = AC = 13 \text{ cm}$ and $BC = 24 \text{ cm}$
 $a = c = 13 \text{ cm}$ & $b = 24 \text{ cm}$

$$s = \frac{a+b+c}{2} = \frac{13+24+13}{2} = \frac{50}{2} = 25 \quad \therefore \boxed{s = 25 \text{ cm}}$$

→ Area of isosceles triangle =

$$\begin{aligned} A &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{25(25-13)(25-24)(25-13)} \\ &= \sqrt{25(12)(1)(12)} \\ &= 12 \times 5 \end{aligned}$$

$$\boxed{A = 60 \text{ Sq. cm}}$$

3. Find Volume of a cylinder whose radius is 5 cm. and height is 12 cm. ($\pi = 3.14$).

→ $r = 5 \text{ cm}$, $h = 12 \text{ cm}$.

$$\begin{aligned} \text{Volume of cylinder } V &= \pi r^2 h \\ &= 3.14 \times 5 \times 5 \times 12 \\ &= \frac{314}{10} \times \frac{5}{2} \times \frac{6}{1} \\ &= 314 \times 30 \end{aligned}$$

$$\boxed{V = 9420 \text{ cm}^3}$$

4. Find Volume of a cone whose diameter is 20 cm. and altitude is 15 cm.

→ Here $d = 20 \text{ cm}$ $h = 15 \text{ cm}$.

$$\therefore r = \frac{d}{2} = \frac{20}{2} = 10 \text{ cm.}$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times 3.14 \times 10 \times 10 \times 15$$

$$\boxed{V = 1570 \text{ cm}^3}$$

6. The side of a rectangular plot are in the ratio 3:2, if the area of the plot is 600 sq. met. Then find the perimeter of the plot.

$$\rightarrow \text{Here } 3x \times 2x = 600$$

$$\therefore 6x^2 = 600$$

$$\therefore x^2 = 100$$

$$\therefore x = 10.$$

Here one side is $3x = 3 \times 10 = 30 \text{ m}$

and another side is $2x = 2 \times 10 = 20 \text{ m}$

Now perimeter of rectangular plot

$$= 2(l+b)$$

$$= 2(30+20)$$

$$= 2(50)$$

$$= 100 \text{ m}$$

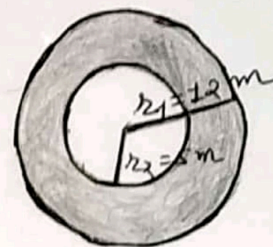
$$\therefore \boxed{P = 100 \text{ m}}$$

7. A circular garden has radius 12 met. A path of 7 met. wide has to be constructed on side periphery of the garden. Find cost of path at the rate of Rs 50 per m^2 .

→ Here $r_1 = 12$.

$$r_2 = 12 - 7 = 5.$$

$$\begin{aligned} \text{Area of path} &= \pi (r_1^2 - r_2^2) \\ &= 3.14 (144 - 25) \\ &= 3.14 (119) \\ &= 373.5 \text{ m}^2. \end{aligned}$$



cost of path at the rate of Rs 50 per m^2

∴

cost of path of area 373.5 m^2 is

$$373.5 \times 50 = 18,675 \text{ Rs.}$$

8. A cylinder is made from a cone with radius 35 cm and height 12 cm. If the radius of cylinder is 7 cm then find the height.

→ Here Radius of cone $r_1 = 35 \text{ cm}$ & $h_1 = 12 \text{ cm}$ and radius of cylinder $r_2 = 7 \text{ cm}$, $h_2 = (?)$

Given that

Volume of cone = Volume of cylinder.

$$\therefore \frac{1}{3} \pi r_1^2 h_1 = \pi r_2^2 h_2$$

$$\therefore \frac{1}{3} \times \frac{5}{35} \times \frac{5}{35} \times 14 = 7 \times 7 \times h_2$$

$$\therefore \boxed{h_2 = 100 \text{ cm}}$$

$\therefore$ So, height of cylinder is $h_2 = 100 \text{ cm}$.

9. Find the surface area and volume of cylinder with 7 cm radius and 21 cm height.

→ Here $r = 7 \text{ cm}$, $h = 21 \text{ cm}$.

→ surface area of cylinder $= 2\pi r(h+r)$

$$= 2 \times 3.14 \times 7(21+7)$$

$$= 14 \times 3.14(28)$$

$$= 1230.56 \text{ cm}.$$

→ Volume of cylinder $V = \pi r^2 h$

$$= 3.14 \times 7 \times 7 \times 21$$

$$\therefore \boxed{V = 3231 \text{ cm}^3}$$

10. How many spherical balls of 2 cm radius can be made from cube of length 44 cm.

→ No of spherical balls $= \frac{\text{Volume of cube}}{\text{Volume of sphere}}$

$$= \frac{L^3}{\frac{4}{3}\pi r^3} = \frac{3 \times (44)^3 \times 7}{4 \times 22 \times 8}$$

$$= 2541.$$

* Question for 4 Marks :-

1. The sides of the triangle are in the ratio 3:4:5 and perimeter of triangle is 120 cm. then find the area of triangle.

→ Here $3x + 4x + 5x = 120$

$$\therefore 12x = 120$$

$$\therefore x = \frac{120}{12}$$

$$\therefore x = 10 \text{ cm.}$$

Now, sides of Triangle is measure of

$$a = 3x = 3 \times 10 = 30 \text{ cm}$$

$$b = 4x = 4 \times 10 = 40 \text{ cm}$$

$$c = 5x = 5 \times 10 = 50 \text{ cm.}$$

$$\text{Now, } s = \frac{a+b+c}{2} = \frac{30+40+50}{2} = \frac{120}{2} = 60$$

Area of Triangle

$$A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{60(60-30)(60-40)(60-50)}$$

$$= \sqrt{60 \times 30 \times 20 \times 10}$$

$$= \sqrt{60 \times 60 \times 10}$$

$$\boxed{A = 60\sqrt{10} \text{ cm}^2}$$

2. If area of circle is 154 sq.m. Then find its circumference
Find total expense of fencing on the circle at the rate of Rs. 200 per meter.

→ Area of circle = 154

$$\therefore \pi r^2 = 154$$

$$\therefore r^2 = \frac{154 \times 7}{22}$$

$$\therefore r^2 = 49$$

$$\therefore \boxed{r = 7 \text{ m.}}$$

→ circumference of circle = $2\pi r$

$$= 2 \times \frac{22}{7} \times 7$$

$$= 44 \text{ m.}$$

Total expense of fencing on the circle at the rate of Rs 200 per meter.

Then cost of fencing is $44 \times 200 = 8800 \text{ Rs.}$

3. A tank for a petrol pump has a capacity 38500 liter and the diameter of tank is 3.5 meters find the height of the tank.

→ 38500 liter

$$1000 \text{ liter} = 1 \text{ m}^3$$

$$\therefore 38500 \text{ liter} = 38.5 \text{ m}^3$$

$$d = 3.5 \text{ m.} \quad r = \frac{d}{2} = \frac{3.5}{2} = \frac{35}{20}$$

$$\text{Volume of cylinder } V = \pi r^2 h$$

$$\therefore 38.5 = \frac{22}{7} \times \frac{35}{20} \times \frac{35}{20} \times h$$

$$\therefore h = \frac{38.5 \times 8}{77}$$

$$\therefore \boxed{h = 4 \text{ met}}$$

4. A cylindrical tank of radius 8 m and height 10 m is filled full of oil. If the oil is transferred to cuboids tank of length 15 m and width 14 m. Find the height of oil in the cuboids tank.

$$\rightarrow r_1 = 8 \text{ m}, h_1 = 10 \text{ m}$$

$$L = 15 \text{ m} \quad b = 14 \text{ m}.$$

Volume of cylinder = Volume of cuboid.

$$\therefore \pi r_1^2 h_1 = L b h$$

$$\therefore \frac{22}{7} \times 8 \times 8 \times 10 = 15 \times 14 \times h$$

$$\therefore h = \frac{22 \times 8 \times 8 \times 10}{7 \times 15 \times 14}$$

$$\therefore h = \frac{22 \times 64}{49 \times 3}$$

$$\therefore \boxed{h = 10 \text{ m}}$$

5. A hemisphere of radius 3 cm is mounted on cylinder of same radius and 10 cm height. Find volume and surface area of the shape.

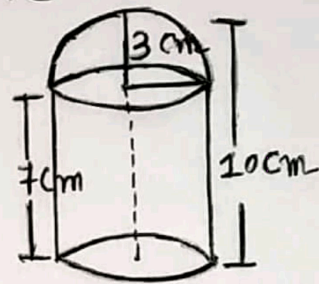
$$\rightarrow h = 10 \text{ cm}, h_1 = 7 \text{ cm}, r = 3 \text{ cm}.$$

→ surface area of cylinder

+ surface area of Hemisphere
= surface area of shape

$$\therefore 2\pi r(h_1 + r) + 3\pi r^2$$

= surface area of shape.



$$\therefore 2\pi 3(7+3) + 3\pi 9 = "$$

$$\therefore 60\pi + 27\pi = \text{surface area of shape.}$$

$$\therefore \text{surface area of shape} = 87\pi \text{ cm}^2.$$

→ Volume of shape = Volume of cylinder
+ Volume of Hemisphere.

$$= \pi r^2 h_1 + \frac{2}{3} \pi r^3$$

$$= \pi 9 \times 7 + \frac{2}{3} \pi 27$$

$$= 63\pi + 18\pi$$

$$= 81\pi \text{ cm}^3$$

6. Cuboid of 15 cm length and 10 cm width is made from cylinders of radius 5 cm and height 20 cm. what is the height of cuboids?

→ Here $l = 15 \text{ cm}$, $b = 10 \text{ cm}$, $h = (?)$
and $r = 5 \text{ cm}$ $h_1 = 20 \text{ cm}$.

→ Volume of cuboid = Volume of cylinder

$$\therefore lbh = \pi r^2 h_1$$

$$\therefore 15 \times 10 \times h = \pi \times 5 \times 5 \times 20$$

$$\therefore h = \frac{5^2 \times 20 \pi}{15 \times 10}$$

$$\therefore \boxed{h = \frac{10\pi}{3} \text{ cm.}}$$

7. A cone of radius 7 cm, slant height 8 cm is mounted on cylinder of 7 cm radius and 12 cm height. Find surface area of the shape. Also find height of cone.

→ Here $r = 7 \text{ cm}$

$$l = 8 \text{ cm}$$

$$h = 12 \text{ cm}$$

$$h_1 = (?)$$

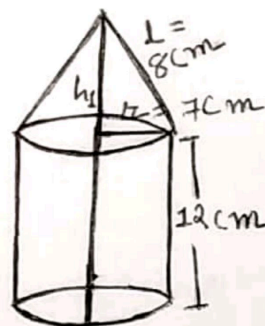
by Pythagoras Rule

$$l^2 = r^2 + h_1^2$$

$$\therefore h_1^2 = l^2 - r^2$$

$$\therefore = 64 - 49$$

$$h_1^2 = 15 \text{ cm} \quad \therefore h_1 = \sqrt{15} \text{ cm.}$$



→ surface area of the shape = surface area of cone + surface area of cylinder

$$= \pi r(l + r) + 2\pi r(h + r)$$

Surface area of the shape

$$= \pi(7)(8+7) + 2\pi(7)(12+7)$$

$$= 7\pi(15) + 14\pi(19)$$

$$= 105\pi + 266\pi$$

$$= 371\pi \text{ cm}^2$$